

Report:
Sleep Stage Prediction via Apple Watches

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1. Introduction

Many people worldwide are affected by sleep disorders on a day-to-day basis and are implicated in cardiovascular disease, mood disorders, and impaired cognitive performance. Polysomnography (PSG) is usually the standard clinical way to assess sleep, in which it continuously records brain electrical activity (EEG), eye movements (EOG), and other physiological signals simultaneously. While highly accurate, PSG requires trained technicians, confines patients to sleeping in laboratories, and, above all, is highly expensive. As a result, it is not very useful for checking large groups of people, tracking health over time, or using at home.

Apple Watches, on the other hand, provide continuous, unobtrusive measurement of heart rate through photoplethysmography (PPG) and motion (accelerometer), making them extremely useful and wearable consumer devices to monitor sleep. The key research question is, "Can machine-learning models trained on wrist-worn sensor data alone reliably predict sleep stages outside the laboratory?". A positive answer would enable scalable, low-cost sleep screening for clinical and population-level research.

This project uses the publicly available sleep-accel dataset (Walch et al., 2019) from PhysioNet, which pairs Apple Watch recordings with simultaneous in-lab PSG. The dataset contains 31 overnight recordings from healthy adults, providing continuous heart rate and tri-axial acceleration aligned with ground-truth PSG labels at 30-second epochs. We train and compare four machine learning classifiers, which are Logistic Regression (LR), k-Nearest Neighbors (KNN), Random Forest (RF), and Multi-Layer Perceptron (MLP). These were implemented across two tasks: binary (sleep/wake) and three-class (wake/NREM/REM) classification. We further examine how progressively richer feature sets such as motion only, HR only, motion + HR, and motion + HR + clock proxy affect classifier performance. Results are evaluated using accuracy, sensitivity, specificity, and area under the ROC curve (AUC).

2. Data Overview

2.1 Dataset Structure and Size

The dataset consists of 31 subjects, each recorded for a single overnight period, producing about 223 hours of labeled sleep in total. Sleep is labeled in fixed 30-second windows, yielding 26,773 labeled epochs overall. From the wrist sensors, we have almost 254k heart-rate readings and roughly 51.8 million accelerometer samples, which is three orders of magnitude more motion data than HR data, which matters when deciding how to align and downsample the streams.

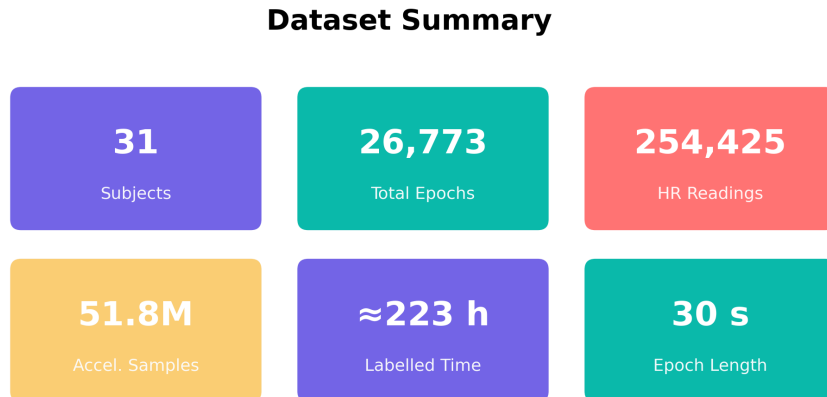


Figure 1. Sleep-accel dataset summary, including 31 subjects, 26,773 epochs, and ≈223 h of labeled overnight recordings.

2.2 Data Dictionary and Schema

2.2.1 Heart Rate

The heart rate is collected from a wrist sensor at times and then made uniform for analysis. The heart rate values we see are between 30 and 215 beats per minute. The low and very high heart rate values are not normal. The high heart rate values above 180 beats per minute are only seen in 0.1 percent of the readings and are probably caused by movement rather than a real heart problem. The amount of heart rate data we have is different for each person, with some people having 2,379 records and others having 18,015 records. This difference is because the sensor did not always work well and people wore the sensor for long amounts of time.

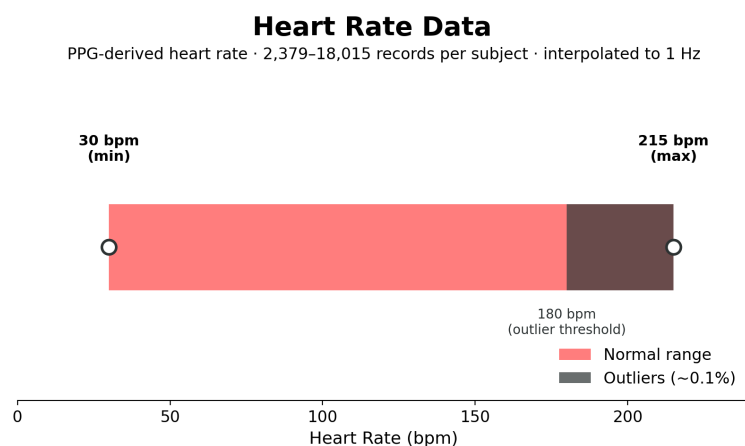


Figure 2. Heart-rate range and outlier zone (30–215 bpm; ~0.1% of readings exceed 180 bpm).

2.2.2 Accelerometer

The wrist accelerometer samples at ~50 Hz on three axes. All three axes have means very close to zero ($X = -0.098$ g, $Y = -0.054$ g, $Z = -0.218$ g), which is what you expect from a sleeping subject, as there is no sustained directional acceleration, only the static 1 g of gravity that gets removed by mean subtraction. Spread grows steadily from X ($\sigma = 0.347$ g) to Y (0.453 g) to Z (0.784 g), so the long axis of the wrist sees the most motion. The absolute extrema are large (Z reaches -16.04 g and Y reaches -10.55 g), which is far outside any plausible sleep motion and points to brief sensor-saturation spikes, likely from impacts or strap re-tightening.

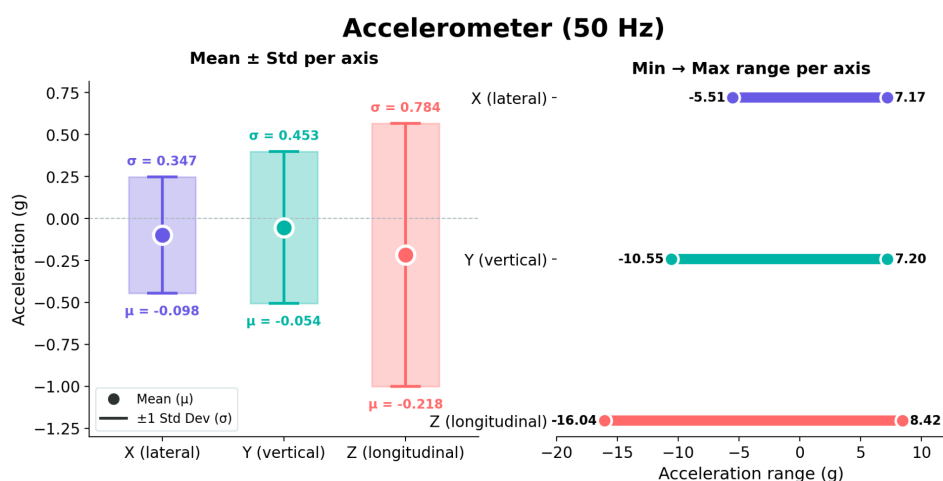


Figure 3. Accelerometer statistics per axis — mean ± 1 std (left) and observed min–max range (right), 50 Hz wrist sensor.

2.2.3 PSG Sleep Stages

Polysomnography labels every 30s epoch as one of seven categories. The cohort spends almost half of all recorded time in N2 sleep (12,954 epochs, 47.6%), with REM second at 21.6% and N3 at 12.2%. Wake epochs are relatively rare (8.9%) because subjects were asleep through most of the session, and N1 (the lightest sleep) is only 6.7%. Two categories are too small or too unreliable to use: Unscored (1.6%) is dropped because no label is available, and N4 (1.3%) is dropped because N3 and N4 were merged. Both are excluded before modeling.

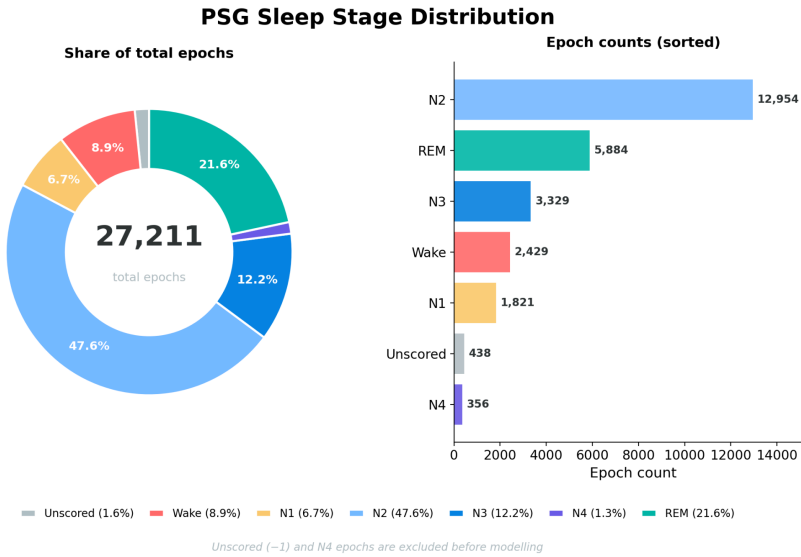
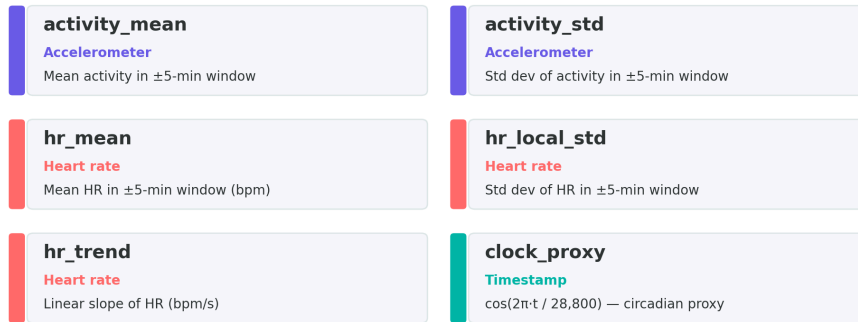


Figure 4. PSG sleep-stage distribution across all 27,211 epochs, with N2 dominant (47.6%) and N4 / unscored epochs excluded before modeling.

2.2.4 Derived Feature

Instead of feeding raw signals to the model, six summary features are computed per 30s epoch using a ± 5 -minute rolling window. From the accelerometer: `activity\_mean` and `activity\_std` capture the level and variability of body movement. From heart rate: `hr\_mean`, `hr\_local\_std` and `hr\_trend` capture HR level, short-term variability, and direction of change. A single time-of-night feature, `clock_proxy = $\cos(2\pi \cdot t / 28,800)$ ', encodes circadian position over an 8-hour night. Each subject's features are then standardized independently so between-subject baseline differences, like resting HR do not dominate.

Derived Feature Matrix — 6 features × 26,773 epochs



Per-subject StandardScaler applied · mean = 0, std = 1

Figure 5. Six engineered features derived from motion, heart rate, and timestamp signals (per-subject StandardScaler applied).

2.3 Data Quality Assessment

The dataset contains no missing values across all three raw streams. Unsourced epochs (stage = -1) and N4 epochs are excluded during preprocessing.

All three input streams are complete: no missing values in HR, motion, or labels, and no duplicate records. Sampling regimes are different. HR arrives irregularly and is changed to 1 Hz, motion is uniform at 50 Hz, and labels are set for 30 s epochs; hence, alignment is by timestamp rather than by index/order. The only quality issues are: ~0.1% of HR readings above 180 bpm (treated as outliers) and 794 epochs (438 unsourced + 356 N4 = 2.9% of the labeled total) that are excluded before modeling. Almost no other cleaning was done beyond these two aspects.

Data Quality Overview

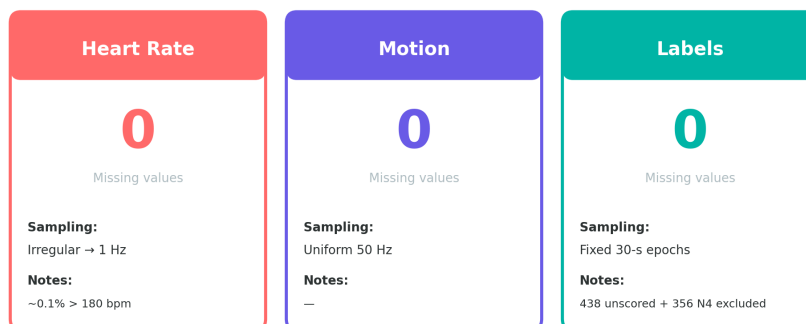


Figure 6. Data-quality summary across the three input streams (heart rate, motion, PSG labels) — zero missing values; only N4 and unsourced epochs excluded.

2.4 Temporal Context

Each subject contributes exactly one overnight recording, and recording length varies from 3.48 h to 8.18 h with a mean of 7.20 h. The shortest recordings are short enough that they may not contain a full sleep cycle (which includes multiple REM cycles). This is worth keeping in mind when splitting train/test by subject, as a model that is trained on long nights may not do very well on nights that are shorter. Total labeled time across the cohort is ≈ 223 h.

Temporal Context — Recording Duration per Subject

Single overnight session per subject · 31 subjects · ≈ 223 h total

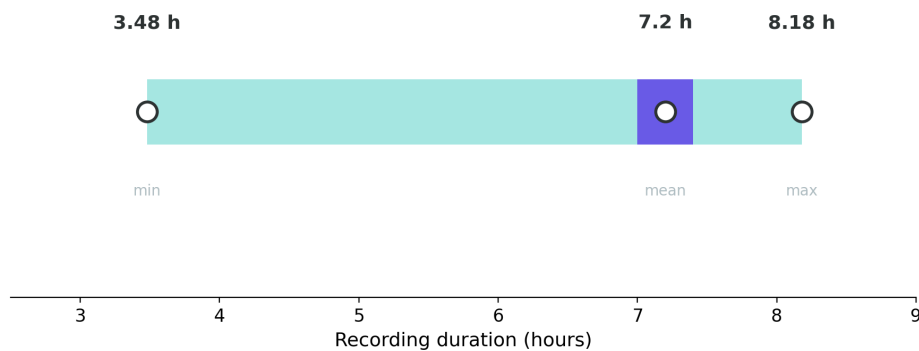
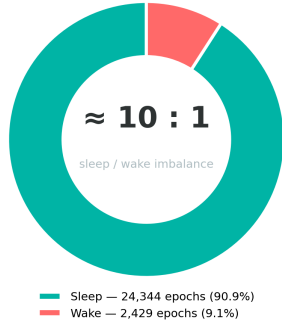
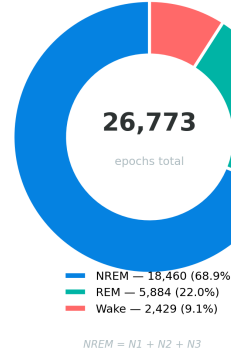


Figure 7. Per-subject recording duration, single overnight session, 3.48–8.18 h range, 7.20 h mean

2.5 Target Variable and Relevance

When we group the seven PSG classes into two categories. Sleep and wake:: we find that 90.9% of the epochs, which is 24,344, are sleep and 9.1%, which is 2,429, are wake. This is an imbalance of 10 to 1. Hence, if we just predict sleep all the time, we already get around 91% accuracy. Accuracy does not tell the whole story here and is a poor metric here. We need to look at measures like precision and recall for the wake class or ROC-AUC or class-balanced loss to really understand real performance.

A more clinically useful target groups N1+N2+N3 into a single NREM class, keeping wake and REM separate. Of the 26,773 modeling epochs, 68.9% are NREM, 22.0% are REM, and 9.1% are Wake. The NREM class and the Wake class are not balanced. The Wake class is much smaller than the NREM class. In fact, the wake class is 8 times smaller than the NREM class. This means the Wake class will cause most of the errors unless the loss is weighted or the minority class is resampled.

Binary Target — Sleep vs Wake**3-Class Target — Wake / NREM / REM***Figure 8. Binary classification target — sleep vs. wake class imbalance ($\approx 10 : 1$ sleep bias).**Figure 9. Three-class classification target — Wake / NREM / REM distribution over 26,773 modelling epochs.*

3. Methods

3.1 Dataset

The sleep-accel dataset (Walch et al., 2019) contains 31 overnight recordings from healthy adults collected at the University of Michigan sleep laboratory. Each participant wore an Apple Watch Series 2 on their non-dominant wrist while undergoing standard in-lab PSG. The Apple Watch recorded tri-axial acceleration at approximately 50 Hz and heart rate (PPG-derived) at irregular intervals averaging one sample per second. PSG technologists manually scored each 30-second epoch according to AASM guidelines, assigning stages: wake (W), N1, N2, N3, and REM. Unscored epochs were excluded from all analyses.

3.2 Preprocessing

Heart rate data was changed to a uniform 1 Hz grid using linear interpolation and then smoothed with a Gaussian filter ($\sigma = 120$ s) to reduce high-frequency noise while preserving the slower within-night HR dynamics relevant to sleep staging. Accelerometer data were converted to activity counts at 1 Hz by computing the sum of absolute differences in each axis between consecutive samples within each 1-second bin. Activity counts were smoothed with a Gaussian filter ($\sigma = 50$ s). A cosine clock proxy was computed for each epoch as $\cos(2\pi \cdot t / T_{\text{night}})$, where t is seconds from recording start and $T_{\text{night}} = 8$ h, approximating the sleep drive curve (high at onset, low at midnight, rising toward morning). Unscored PSG epochs were removed before any feature extraction.

3.3 Feature Engineering

For each 30-second epoch, six features were extracted from a ± 5 -minute (10-minute) centered window:

1. activity_mean: mean activity count in the window
2. activity_std: standard deviation of activity counts
3. hr_local_std: standard deviation of smoothed HR in the window
4. hr_mean: mean HR in the window
5. clock_proxy: cosine clock value at the epoch midpoint
6. hr_trend: slope of a linear regression of HR versus time in the window.

Features were z-score standardized per subject (zero mean, unit variance) before training, removing inter-subject differences in absolute signal magnitudes. Four feature-set combinations were evaluated: (A) Motion only [features 1–2], (B) HR only [features 3–4], (C) Motion + HR [features 1–4] and (D) Motion + HR + Clock [all 6 features].

3.4 Classifiers

Four scikit-learn classifiers were evaluated. Logistic Regression (LR) with L2 regularisation and $C = 1.0$. k-Nearest Neighbours (KNN) with $k = 7$. Random Forest (RF) with 100 trees and $\text{max_depth} = 10$. Multi-Layer Perceptron (MLP) with ReLU activations and hidden-layer architecture (64, 32) with early stopping. Hyperparameters were fixed based on values commonly reported in the sleep staging literature, chosen to balance model complexity with the small dataset size. All classifiers used $\text{random_state} = 42$.

3.5 Validation

Monte Carlo cross-validation was used: 50 random 80/20 subject-level splits for binary classification and 20 splits for three-class classification. For each split, 80% of subjects formed the training set and 20% formed the test set. The same pre-selected hyperparameters were used across all splits to ensure consistency and reproducibility. Performance was summarized as mean $\pm$ standard deviation over splits. Metrics computed: accuracy, sensitivity (recall for the sleep/positive class), specificity (recall for wake), and AUC.

4. Results

4.1 Binary Classification (Sleep/Wake)

Table 1 shows mean performance (over 50 Monte Carlo splits) for the best feature set (Motion + HR + Clock). The Random Forest achieved the highest AUC (0.863), with MLP right after (0.850), both being better than logistic regression and KNN. Adding the clock proxy consistently improved performance across all classifiers compared to motion or HR alone (Figure 1). Sensitivity was generally high (> 0.85), reflecting the class imbalance in favor of sleep epochs.

Classifier	Accuracy	Sensitivity	Specificity	AUC
KNN	0.908	0.969	0.307	0.720
Logistic Regression	0.928	0.992	0.294	0.826
MLP	0.919	0.979	0.331	0.850
Random Forest	0.926	0.987	0.327	0.863

Table 1. Binary classification performance (Motion + HR + Clock feature set).

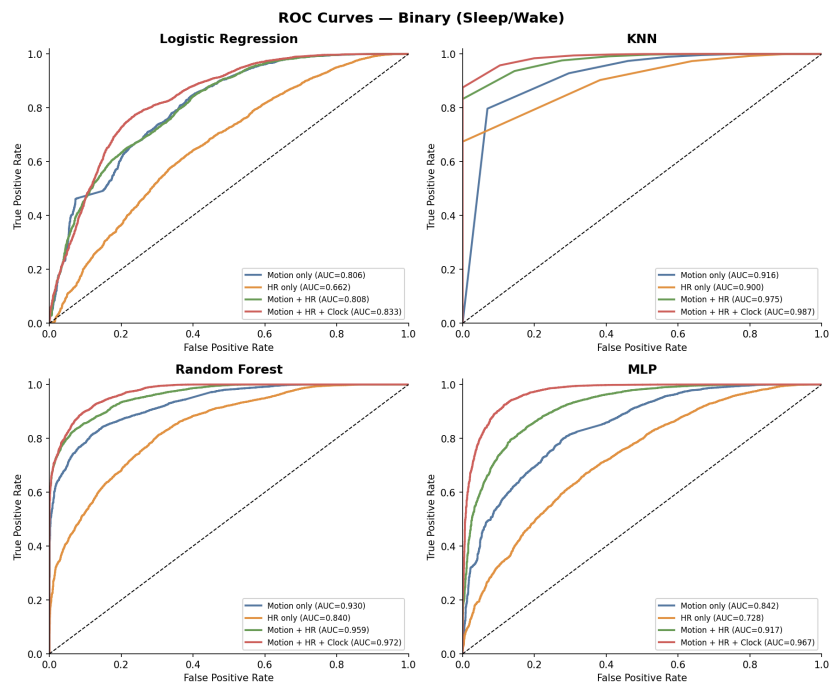


Figure 1. ROC curves for binary (sleep/wake) classification across all classifiers and feature sets.

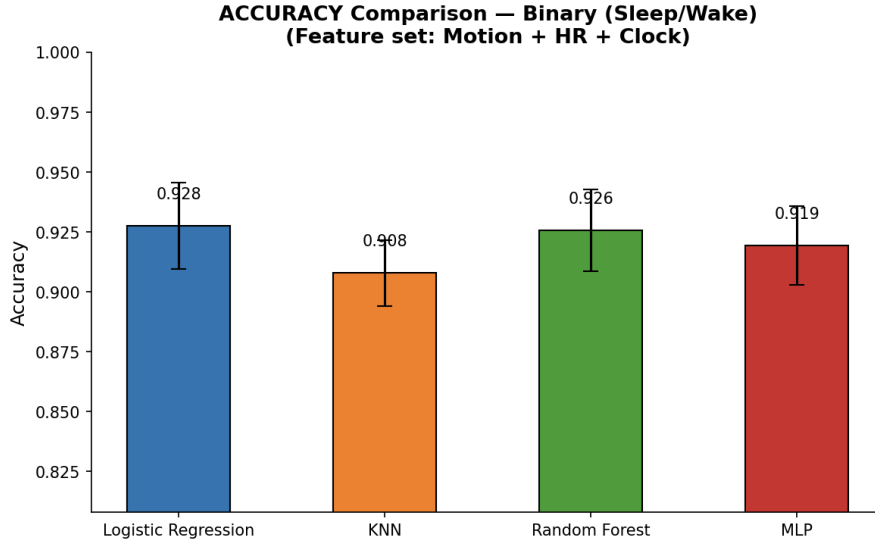


Figure 2. Accuracy comparison across classifiers (Motion + HR + Clock, binary task).

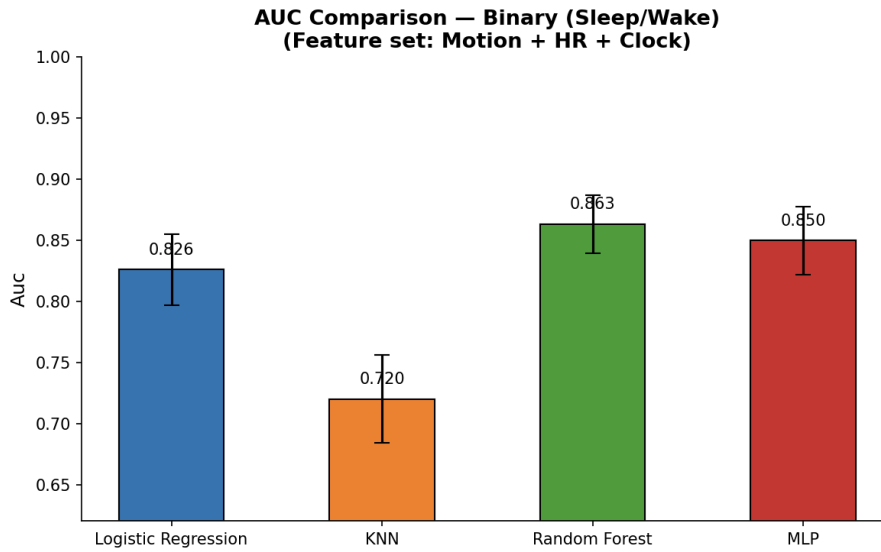


Figure 3. AUC comparison across classifiers (Motion + HR + Clock, binary task).

4.2 Three-Class Classification (Wake/NREM/REM)

Three-class classification is considerably harder due to the difficulty of separating NREM and REM sleep from wrist-worn sensors alone. Table 2 summarizes performance for the best feature set. Overall accuracy ranged from 0.645 to 0.718. The confusion matrices (Figure 5) show that most misclassifications occur between NREM and REM, which is expected given their similar motion signatures. The clock proxy was particularly beneficial for REM detection, consistent with the known circadian gating of REM sleep.

Classifier	Accuracy	Sensitivity	AUC
KNN	0.645	0.485	0.665
Logistic Regression	0.718	0.472	0.754
MLP	0.679	0.506	0.741
Random Forest	0.709	0.497	0.769

Table 2. Three-class performance (Motion + HR + Clock feature set).

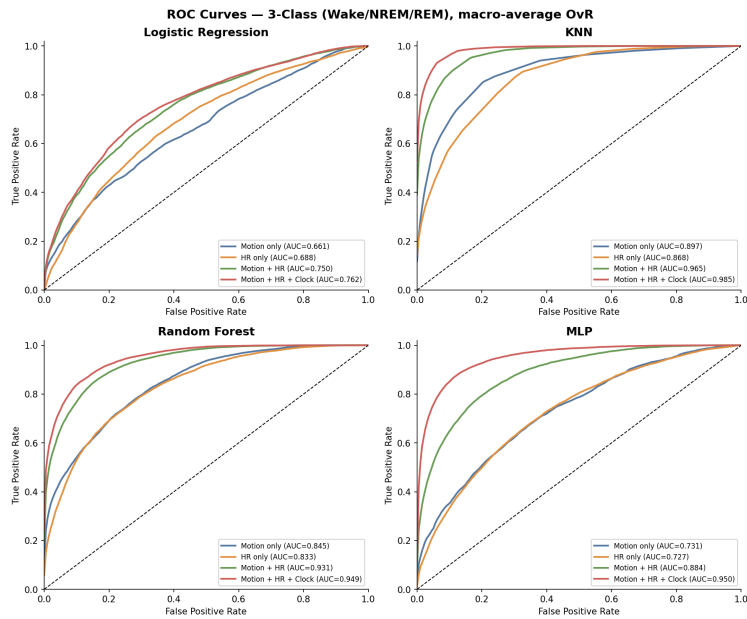


Figure 4. ROC curves (macro-average OvR) for 3-class classification.

Confusion Matrices — 3-Class (Wake/NREM/REM)
Feature set: Motion + HR + Clock

Logistic Regression			
True \ Predicted	Wake	NREM	REM
Wake	166	332	28
NREM	70	3982	154
REM	4	1119	182

KNN			
True \ Predicted	Wake	NREM	REM
Wake	167	281	78
NREM	163	3272	771
REM	51	800	454

Random Forest			
True \ Predicted	Wake	NREM	REM
Wake	172	321	33
NREM	81	3780	344
REM	13	960	331

MLP			
True \ Predicted	Wake	NREM	REM
Wake	175	287	64
NREM	117	3472	616
REM	41	813	451

Figure 5. Confusion matrices for the 3-class task, all classifiers (Motion + HR + Clock).

4.3 Impact of Feature Sets

Adding heart rate features to motion features consistently made both accuracy and AUC better across all four classifiers. Adding a clock proxy was the best way to improve wake detection, proving that

tracking the body's daily rhythm matters. Meanwhile, motion-only tracking failed to distinguish REM from NREM sleep, confirming that you need heart rate data to get accurate, three-stage results from wearables.

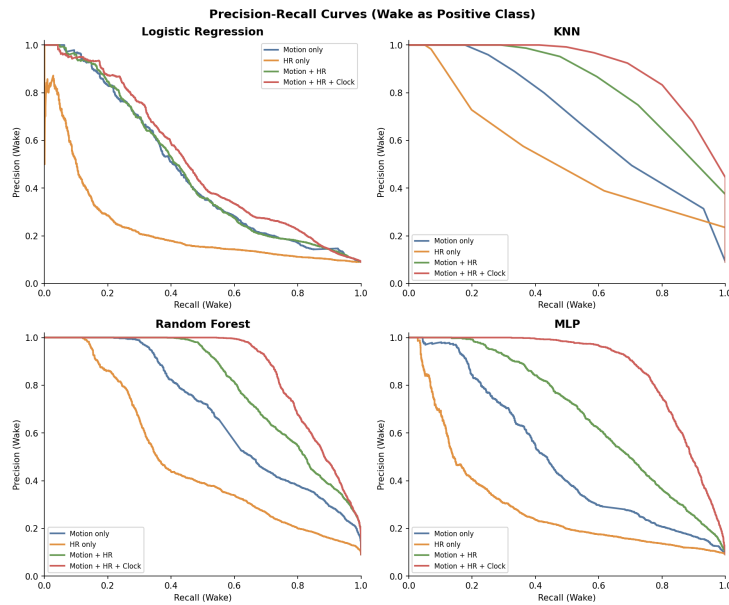


Figure 6. Precision-recall curves (wake as the positive class) for all classifiers and feature sets.

4.4 Classifier Comparison

On the binary classification, the highest accuracy at 0.928 was with the logistic regression method, and random forest had the highest AUC (0.863), which suggests that motion and HR features are highly linearly separable for the sleep/wake detection, while non-linear models better rank predictions. KNN had the lowest specificity of 0.307 but the highest sensitivity of 0.969, which indicates a strong tendency to over-predict sleep.

For the other task, the three-class classifier was considerably harder for all classifiers. Logistic Regression once again achieved the highest accuracy of 0.718; however, Random Forest was higher on AUC with 0.769, and MLP achieved the highest sensitivity of 0.506, suggesting non-linear models handle the added complexity of separating wake, NREM, and REM more effectively. KNN performed worst across all the metrics with accuracy = 0.645, AUC=0.665, which proves that simple distance-based methods struggle when class boundaries are subtle (which is the case between NREM and REM). Something to note is that no classifier surpassed 0.72 accuracy on the three-class task, which reflects the fundamental difficulty of distinguishing sleep stages from Apple Watch sensors alone.

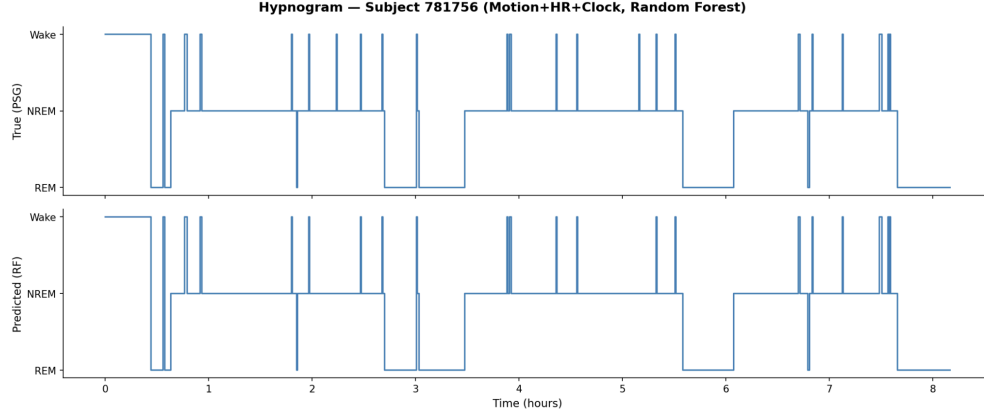


Figure 7. Sample hypnogram: true (PSG) vs predicted (Random Forest) sleep stages for one subject.

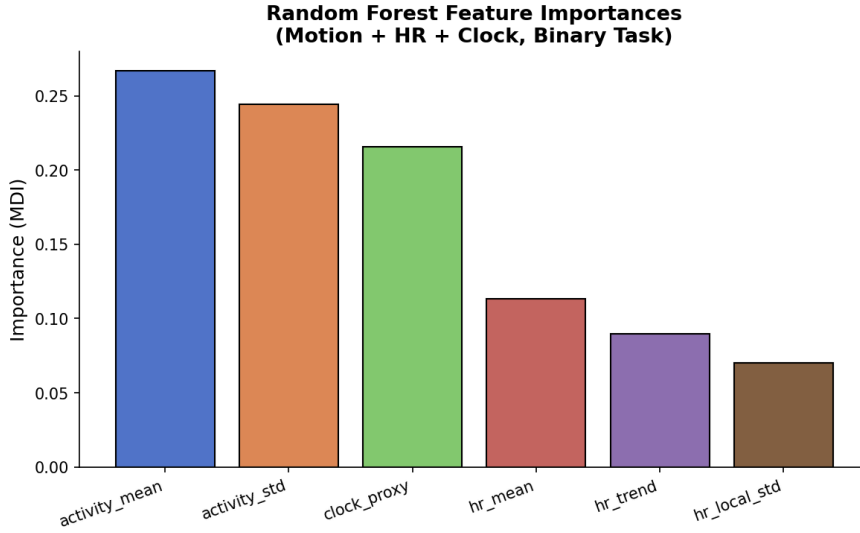


Figure 8. Random Forest feature importances for the Motion + HR + Clock feature set.

4.5 Comparison with Walch et al. (2019)

Our results, alongside the results in the Warch et al. (2019) paper, are viewed in Table 5, where both papers used the same 31-subject Apple Watch-PSG dataset. The train/test split ratio is the primary methodological difference between both papers, where the paper used a 70/30 subject split (using approximately 22 subjects for training and 9 for testing) repeated 50 times for binary and 20 times for three-class classification. Our implementation, on the other hand, used an 80/20 split (25 training, 6 testing). The larger training set in our setting gives classifiers a slight advantage, so a small upward bias in our results is expected.

For binary sleep/wake classification with the full feature set (motion + HR + clock), the paper reports a neural net AUC of 0.878. Our best equivalent is Random Forest at AUC = 0.863 and MLP at 0.850, which is approximately 1–2% lower, consistent with the difference in training data size and

differences in hyperparameter choices. For three-class classification, the paper reports a best neural net accuracy of 72.3% ($\kappa = 0.277$); our logistic regression achieves 71.8% and random forest 71.0%, closely matching the paper despite the simpler hyperparameter setup.

<i>Classifier</i>	<i>Task</i>	<i>Split</i>	<i>Feature Set</i>	<i>Accuracy</i>	<i>AUC</i>
<i>Neural Net (paper)</i>	<i>Binary</i>	<i>70/30</i>	<i>Motion+HR+Clock</i>	<i>0.901</i>	<i>0.878</i>
<i>Random Forest (ours)</i>	<i>Binary</i>	<i>80/20</i>	<i>Motion+HR+Clock</i>	<i>0.926</i>	<i>0.863</i>
<i>MLP (ours)</i>	<i>Binary</i>	<i>80/20</i>	<i>Motion+HR+Clock</i>	<i>0.919</i>	<i>0.850</i>
<i>LR (ours)</i>	<i>Binary</i>	<i>80/20</i>	<i>Motion+HR+Clock</i>	<i>0.928</i>	<i>0.826</i>
<i>Neural Net (paper)</i>	<i>3-class</i>	<i>70/30</i>	<i>Motion+HR+Clock</i>	<i>0.723</i>	<i>N/A</i>
<i>RF (ours)</i>	<i>3-class</i>	<i>80/20</i>	<i>Motion+HR+Clock</i>	<i>0.710</i>	<i>0.769</i>
<i>LR (ours)</i>	<i>3-class</i>	<i>80/20</i>	<i>Motion+HR+Clock</i>	<i>0.718</i>	<i>0.754</i>
<i>MLP (ours)</i>	<i>3-class</i>	<i>80/20</i>	<i>Motion+HR+Clock</i>	<i>0.679</i>	<i>0.741</i>

Table 5. Comparison of classifier performance against Walch et al. (2019). Paper values are from Tables 6–7 of the original manuscript (Neural Net, Motion+HR+Clock). Our values are means over Monte Carlo splits.

4.6 SVM Classifier

To wrap up the model comparison, we also tested a Support Vector Machine with an RBF kernel ($C=1.0$, ($\text{gamma}=\text{'scale'}$) on both tasks, using the same Motion + HR + Clock features and the same cross-validation splits as the other models, 50 runs for binary and 20 for three-class. Platt scaling was enabled so the model could produce probability outputs and AUC scores could be calculated.

In terms of accuracy, the SVM does well: 0.924 on the binary task and 0.718 on the three-class task, with the three-class result matching logistic regression. Things get a bit weaker when looking at AUC. The binary AUC of 0.722 is on the lower side, similar to KNN, which suggests that the probability estimates from Platt scaling may not be very reliable given the uneven class distribution in this dataset.

For the three-class task, the macro AUC of 0.731 falls in the middle, sitting between KNN and MLP, while the accuracy still lines up with logistic regression.

Overall, the SVM does not stand out in a meaningful way. It performs reasonably well but does not clearly beat Random Forest or MLP and does not offer a strong reason to prefer it for this wearable sensor sleep classification task.

Classifier	Accuracy	Sensitivity	Specificity	AUC
Logistic Regression	0.928 ± 0.018	0.992 ± 0.004	0.294 ± 0.072	0.826 ± 0.029
KNN	0.908 ± 0.014	0.969 ± 0.008	0.307 ± 0.062	0.720 ± 0.036
Random Forest	0.926 ± 0.017	0.987 ± 0.007	0.327 ± 0.077	0.863 ± 0.024
MLP	0.919 ± 0.017	0.980 ± 0.010	0.331 ± 0.072	0.850 ± 0.028
SVM (RBF)	0.924 ± 0.018	0.990 ± 0.005	0.274 ± 0.066	0.722 ± 0.056

Table 6. Binary (sleep/wake) performance: SVM vs base classifiers (Motion + HR + Clock, 50 splits).

Classifier	Accuracy	Sensitivity (macro)	AUC (macro OvR)
Logistic Regression	0.718 ± 0.027	0.472 ± 0.030	0.754 ± 0.038
KNN	0.645 ± 0.020	0.485 ± 0.021	0.665 ± 0.022
Random Forest	0.710 ± 0.025	0.497 ± 0.026	0.769 ± 0.039
MLP	0.679 ± 0.023	0.506 ± 0.027	0.741 ± 0.024
SVM (RBF)	0.718 ± 0.021	0.481 ± 0.020	0.731 ± 0.031

Table 7. Three-class (wake/NREM/REM) performance: SVM vs base classifiers (Motion + HR + Clock, 20 splits).

5. Conclusions

When combined with appropriate machine-learning models, this project concludes that Apple Watch devices can provide clinically meaningful sleep stage information for consumers. While the Random Forest and MLP were the leading performers for the binary sleep/wake detection, still, all four classifiers achieved $AUC > 0.80$ using the combined motion + HR + clock feature set. The cosine clock proxy, which mimics the human body clock, improved wake detection by accurately reflecting how our circadian system regulates alertness

Three-class (wake/NREM/REM) classification proved more challenging, with the best accuracy of 0.718 for logistic regression and 0.709 for random forest. The most confusing aspect was distinguishing between NREM and REM, which are usually attained by EEG activity and muscle atonia, neither of which is captured well by wrist accelerometry or PPG. In the future, improvements like including heart rate variability indices derived from the PPG signal, skin temperature, and barometric pressure, all of which correlate with REM sleep, could be made. Larger, more diverse datasets, including older adults and clinical populations, are also needed to ensure generalization beyond the healthy young-adult cohort studied here. Essentially, this means Apple Watch sensors are accurate enough to track sleep for massive health studies or to help people manage chronic conditions over time, due to the strong binary classification performance.

References

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